TEST REPORT

Report No. :	SHES230601129801
Date of issue	2023-08-29
Test item description:	2D TOF LIDAR
Model/Type reference: :	LoraBeam1, LoraBeam1H, LoraBeam1S, KoraBeam1, KoraBeam1S
Electrical Rating	9V~36Vdc, 2.5w
Applicant	Richbeam (Suzhou) Technology Co., LTD.
Address:	No.788, Southeast Avenue, Changshu high tech industrial Development zone, Jiangsu, China
Manufacturer's Name:	Richbeam (Suzhou) Technology Co., LTD.
Address:	No.788, Southeast Avenue, Changshu high tech industrial Development zone, Jiangsu, China
Testing Laboratory	SGS-CSTC Standards Technical Services (Shanghai) Co., Ltd.
Address	588 West Jindu Road, Xinqiao, Songjiang, 201612 Shanghai, China
Number of Samples received	3
Date of samples reception	2023-07-13
Date Test Conducted	2023-07-26
Test Requested	IP 67
Test Method (standards)	Clause 13.4, 13.5 and 14.2.4 of IEC 60529:1989 + A1:1999 + A2:2013 IP 6 X: Clause 13.4 and 13.6 IP X7: Clause 14.2.7
Test result	Pass
CONCLUSION	The submitted sample **complied** with the Clauses as above

Prepared and checked by:

Nino Sun

Nino Sun

Reviewed by:

Lucy Wang

Unless otherwise agreed in writing, this document is issued by the Company subject to its General Conditions of Service printed overleaf, available on request or accessible at https://www.sgs.com/en/Terms-and-Conditions. Attention is drawn to the limitation of liability, indemnification and jurisdiction issues defined therein. Any holder of this document is advised that information contained hereon reflects the Company's findings at the time of its intervention only and within the limits of Client's instructions, if any. The Company's sole responsibility is to its Client and this document does not exonerate parties to a transaction from exercising all their rights and obligations under the transaction documents. This document cannot be reproduced except in full, without prior written approval of the Company. Any unauthorized alteration, forgery or falsification of the content or appearance of this document is unlawful and offenders may be prosecuted to the fullest extent of the law. Unless otherwise stated the results shown in this test report refer only to the sample(s) tested and such sample(s) are retained for 30 days only.
Attention: To check the authenticity of testing /inspection report & certificate, please contact us at telephone: (86-755) 8307 1443, or email: CN.Doccheck@sgs.com
NO.588 West Jindu Road, Songjiang District, Shanghai, China 201612 t (86-21) 61915666 f (86-21) 61915678 www.sgsgroup.com.cn
中国·上海·松江区金都西路 588 号 邮编: 201612 t (86-21) 61915666 f (86-21) 61915678 sgs.china@sgs.com
Member of the SGS Group (SGS SA)

TEST RESULTS

Standards used IEC 60529:1989 + A1:1999 + A2:2013

Clause	Requirement + Test	Result – Remark	Verdict
13.4, 13.6	IP6X Test	See Observations	Pass
14.2.7	IPX7 Test	See Observations	Pass

TEST PREPARATION:

The samples were mounted in normal operation condition.

TEST CONDITION:

IP6X Test

1. Test means: Dust chamber, with pressure
   The chamber contains 2 kg of powder for every cubic metre of its volume
2. Duration of Test: 2 hours

IPX7 Test

1. Test position: The lowest point of enclosures with a height less than 850 mm is located 1000 mm below the surface of the water
2. Test temperature: Not diver from equipment by more than 5K
3. Duration of Test: 30 min

OBSERVATIONS:

There was no deposit of talcum powder inside the enclosure after IP6X test;

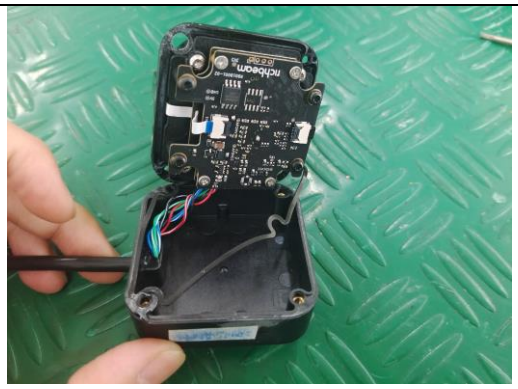
There was no trace of water depositing inside the enclosure after IPX7 test.

Unless otherwise agreed in writing, this document is issued by the Company subject to its General Conditions of Service printed overleaf, available on request or accessible at https://www.sgs.com/en/Terms-and-Conditions. Attention is drawn to the limitation of liability, indemnification and jurisdiction issues defined therein. Any holder of this document is advised that information contained hereon reflects the Company's findings at the time of its intervention only and within the limits of Client's instructions, if any. The Company's sole responsibility is to its Client and this document does not exonerate parties to a transaction from exercising all their rights and obligations under the transaction documents. This document cannot be reproduced except in full, without prior written approval of the Company. Any unauthorized alteration, forgery or falsification of the content or appearance of this document is unlawful and offenders may be prosecuted to the fullest extent of the law. Unless otherwise stated the results shown in this test report refer only to the sample(s) tested and such sample(s) are retained for 30 days only.
Attention: To check the authenticity of testing /inspection report & certificate, please contact us at telephone: (86-755) 8307 1443, or email: CN.Doccheck@sgs.com
NO.588 West Jindu Road, Songjiang District, Shanghai, China 201612 t (86-21) 61915666 f (86-21) 61915678 www.sgsgroup.com.cn
中国·上海·松江区金都西路 588 号 邮编: 201612 t (86-21) 61915666 f (86-21) 61915678 sgs.china@sgs.com
Member of the SGS Group (SGS SA)

Photos of the product:













Unless otherwise agreed in writing, this document is issued by the Company subject to its General Conditions of Service printed overleaf, available on request or accessible at https://www.sgs.com/en/Terms-and-Conditions. Attention is drawn to the limitation of liability, indemnification and jurisdiction issues defined therein. Any holder of this document is advised that information contained hereon reflects the Company's findings at the time of its intervention only and within the limits of Client's instructions, if any. The Company's sole responsibility is to its Client and this document does not exonerate parties to a transaction from exercising all their rights and obligations under the transaction documents. This document cannot be reproduced except in full, without prior written approval of the Company. Any unauthorized alteration, forgery or falsification of the content or appearance of this document is unlawful and offenders may be prosecuted to the fullest extent of the law. Unless otherwise stated the results shown in this test report refer only to the sample(s) tested and such sample(s) are retained for 30 days only.
Attention: To check the authenticity of testing /inspection report & certificate, please contact us at telephone: (86-755) 8307 1443, or email: CN.Doccheck@sgs.com
NO.588 West Jindu Road, Songjiang District, Shanghai, China 201612 t (86-21) 61915666 f (86-21) 61915678 www.sgsgroup.com.cn
中国·上海·松江区金都西路 588 号 邮编: 201612 t (86-21) 61915666 f (86-21) 61915678 sgs.china@sgs.com
Member of the SGS Group (SGS SA)

— End of Report —

Unless otherwise agreed in writing, this document is issued by the Company subject to its General Conditions of Service printed overleaf, available on request or accessible at https://www.sgs.com/en/Terms-and-Conditions. Attention is drawn to the limitation of liability, indemnification and jurisdiction issues defined therein. Any holder of this document is advised that information contained hereon reflects the Company's findings at the time of its intervention only and within the limits of Client's instructions, if any. The Company's sole responsibility is to its Client and this document does not exonerate parties to a transaction from exercising all their rights and obligations under the transaction documents. This document cannot be reproduced except in full, without prior written approval of the Company. Any unauthorized alteration, forgery or falsification of the content or appearance of this document is unlawful and offenders may be prosecuted to the fullest extent of the law. Unless otherwise stated the results shown in this test report refer only to the sample(s) tested and such sample(s) are retained for 30 days only.
Attention: To check the authenticity of testing /inspection report & certificate, please contact us at telephone: (86-755) 8307 1443, or email: CN.Doccheck@sgs.com
NO.588 West Jindu Road, Songjiang District, Shanghai, China 201612 t (86-21) 61915666 f (86-21) 61915678 www.sgsgroup.com.cn
中国·上海·松江区金都西路 588 号 邮编: 201612 t (86-21) 61915666 f (86-21) 61915678 sgs.china@sgs.com
Member of the SGS Group (SGS SA)